CSI 436/536 · PROJECT P · MIDTERM · GROUP 8

# Cheap Text Monitors × Pretrained Representations

**Headline:** do pretrained features help a cheap text monitor? Our first answer was **no** — and it was mostly an artifact of the **input construction**, not the representations.

Three regimes — training size n · distribution shift · perturbation — plus one input ablation that changed the answer.

Jiayu Liao · Zeyu Pan · Yuhan Sun

## The 60-second sprint — everything required, on purpose 

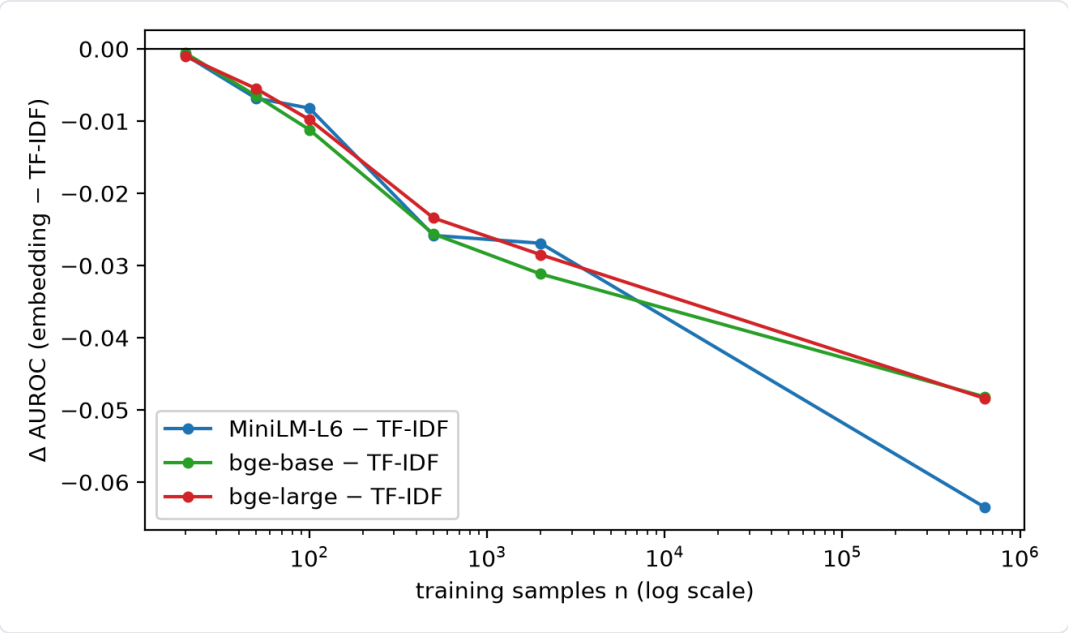
- **Background.** Verifying LLM reasoning step-by-step costs GPT-level calls or human labels — a **cheap text-only monitor** decides: is this step correct?
- **Problem setup.** `Δ(θ) = AUROC(pretrained) − AUROC(TF-IDF)` over regimes θ = (train size n, distribution shift, perturbation — plus an input-construction ablation). **The only variable is where features come from.**
- **Data.** PRM800K (OpenAI, MIT license): **911,867** human-labeled steps, **27.9%** incorrect; split **grouped by problem** — 632,422 train / 88,872 val (unused) / 190,573 test.
- **Methods.** Frozen MiniLM / bge-base / bge-large as feature extractors only; downstream = in-class **logistic regression & linear SVM**; baseline = word+char TF-IDF.
- **Metrics.** Balanced accuracy / AUROC (never raw accuracy); 10 stratified seeds; grouped CV inside train; **test touched once**.

## The 60-second sprint — the scorecard 

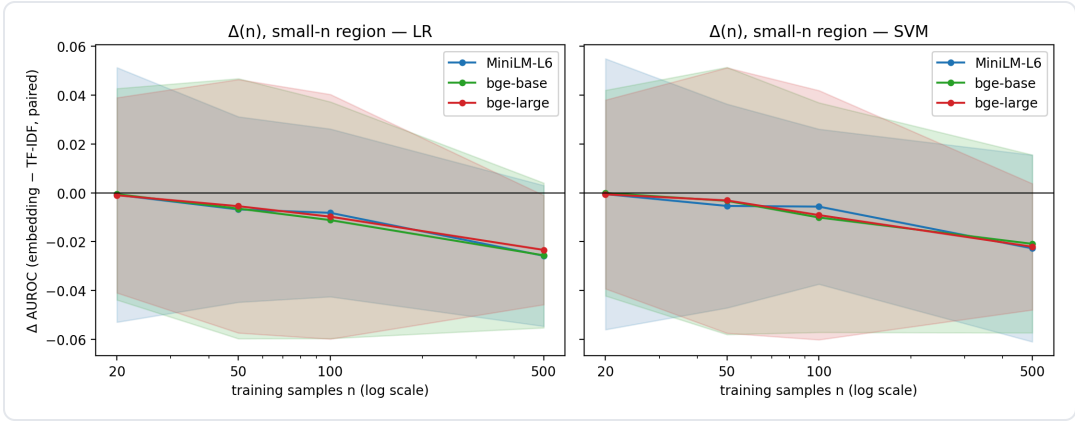
- ✓ **Soundness** — leakage-free grouped split; balanced metrics; 10-seed CIs; hyperparameters by grouped CV only.
- ✓ **Organization** — this checklist first; then one story, told in gaps, not levels.
- ✓ **Clarity** — one figure per claim; every conclusion is a Δ with an interval, never a bare number.
- ✓ **Current progress** — E1–E4 complete on clean v2 data, **plus a reviewer-driven input ablation (E7) that changed the in-domain answer**.
- ✓ **Challenges** — a self-inflicted label bug, caught and fully re-run; a negative headline Δ, **traced to input construction, not hidden**.

> Checklist done. Now the story: a negative result that a reviewer question broke open.

# The first answer — and it didn't look good — Δ(n) in-domain




Δ AUROC (embedding – TF-IDF), input = problem+step · 10 seeds, 95% band




paired Δ at n = 20…500, per-seed

- Textbook expectation: pretrained features help **most when n is small** (sample efficiency).
- Measured (input = problem + step): n = 20 → everything ≈ **0.51, near random**; n ≥ 50 → Δ turns **negative**, monotone, 10/10 seeds same sign.
- Full-data end: **`Δ = −0.045 … −0.064`** (SVM), 95% CI excludes 0 — TF-IDF wins, significantly.
- We reported it as an honest boundary. Then a reviewer asked: **is it the features — or the way we feed them?**

# Autopsy: features, or input?

E7 ablation, reviewer-driven

## Three suspects

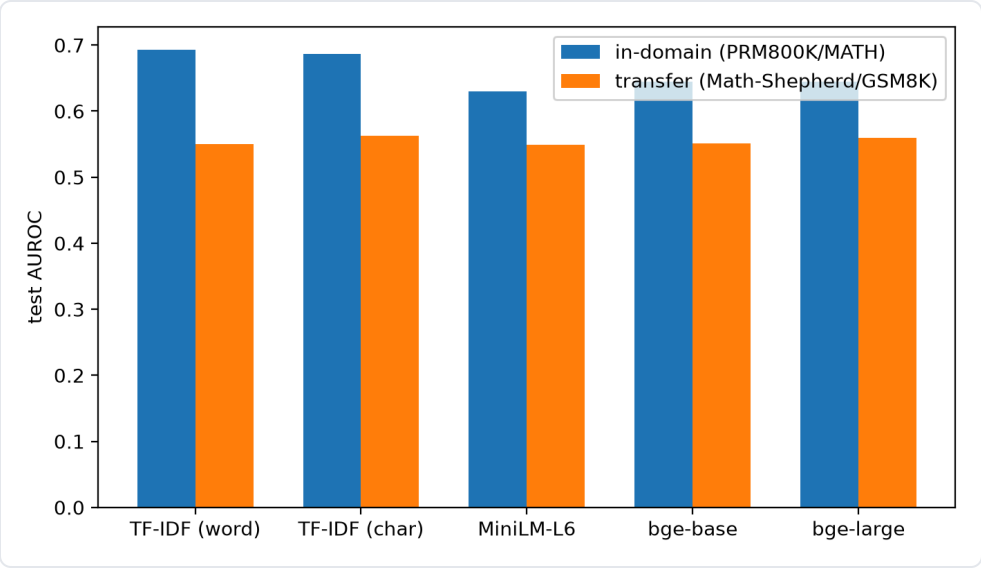
- **Truncation** — median input 89 tokens; MiniLM truncates 1.6% of rows (0.6% lose the whole step), bge 0.04%. Real, but far too small.
- **Position** — moving the step first: embeddings **+0.015…+0.026**, TF-IDF ≈ 0.
- **Dilution** — dropping the problem text: embeddings **+0.03…+0.06**, TF-IDF only +0.012…0.016. The problem text hurts embeddings far more than TF-IDF (consistent with pooling dilution).
- **Verdict** — the in-domain negative Δ was mostly an **input-construction effect, not a feature property**.

## E1 full-data point, test AUROC (SVM)

| input → | step at tail | step first | step only |
|---|---|---|---|
| TF-IDF (word) | 0.696 | 0.694 | 0.708 |
| TF-IDF (char) | 0.687 | 0.687 | 0.704 |
| MiniLM | 0.631 | 0.647 | 0.681 |
| bge-base | 0.650 | 0.668 | 0.704 |
| **bge-large** | 0.651 | 0.677 | **0.711** |

step-only: bge-large **0.711** draws level with TF-IDF **0.708** — Δ goes from −0.045 to ≈ 0 (point estimate).

# The turn, amplified — shift with the fixed input




E3 · MATH → GSM8K, original input (problem+step) · composite shift, reported as such

- **Original input:** bge-large **`Δ = +0.020`** `[0.018, 0.021]`; bge-base +0.006; MiniLM n.s.
- **Step-only input — Δ triples:** bge-large **`+0.069`** `[0.067, 0.071]`; bge-base **`+0.049`** `[0.048, 0.051]`; MiniLM **`+0.030`** `[0.029, 0.032]` — **all three significant**, size gradient intact.
- Removing the problem **hurts** TF-IDF transfer (`0.551 → 0.527`): it carried transferable lexical cues; without them TF-IDF is stuck with MATH-specific surface tokens.
- **E4 perturbation:** parity (≤ 0.007). **E5 selective prediction:** TF-IDF marginally better — reported as-is.

Once the input lets them speak, pretrained features pay everywhere —
and most when the distribution **drifts**.

## Plans → final

- **Open question:** why does step-only input help at all? PRM800K steps may carry their own context — dig in.
- **Cheapest mechanism check:** encode problem and step separately, then concatenate — keeps context, avoids pooling dilution. Plus the step-only Δ(n) curve and in-domain CI (features cached, minutes).
- **Single-factor decomposition** of the compound MATH→GSM8K shift (problem type / source / labeling).
- **E6 (started):** predict downstream Δ from representation properties — ρ = 0.90 on 4 points (preliminary); final = three Δ curves under **both input constructions**, as a measurement report.

> We asked *whether* pretrained representations help. The answer survived a hostile review:
> **yes — if the input lets them speak, and loudest when the distribution drifts.**

# Backup — numbers for Q&A

not timed

| axis θ | comparison | Δ / value | 95% CI | read |
|---|---|---|---|---|
| n (in-domain, original input) | embed – TF-IDF, full data | **–0.045 … –0.064 (SVM)** | excl. 0 | TF-IDF wins — but see E7 |
| input ablation (E7) | bge-large: tail → first → only | 0.651 → 0.677 → 0.711 | — | consistent with dilution |
| step-only, in-domain | bge-large vs TF-IDF(word) | 0.711 vs 0.708 (Δ ≈ +0.004) | point est. | draws level |
| truncation | share of rows truncated | MiniLM 1.6%, bge 0.04% | — | too small to explain the gap |
| shift MATH→GSM8K (original) | bge-large – TF-IDF | **+0.020** | [0.018, 0.021] | sign flips |
| shift (step-only) | bge-large – TF-IDF | **+0.069** | [0.067, 0.071] | triples; all 3 embeds significant |
| overfitting | train–test gap | 0.153–0.168 vs 0.040–0.065 | — | TF-IDF ≈3× gap |
| perturbation | accuracy drop | ≤ 0.007 both | — | parity (honest null) |
| selective pred. | risk @ full coverage | TF-IDF 0.271 | — | TF-IDF marginally better |
| E6 predictor | Spearman ρ | 0.90 (n = 4) | — | preliminary |

- Likely Q: “Why did the first answer say no?” — the step sat at the tail of a long prompt; the problem text hurt embeddings far more than TF-IDF, consistent with pooling dilution (truncation was minor). Step-only closes the in-domain gap.
- Likely Q: “Is the in-domain tie significant?” — Δ ≈ +0.004 is a point estimate; we say “draws level”, not “wins”.
- Likely Q: “Why not split randomly?” — steps of one problem share wording; random split leaks → inflated metrics. We grouped by problem.
- Likely Q: “Why is AUROC not higher overall?” — step-level correctness is genuinely hard; we report Δ between feature sources, not absolute skill.
- Likely Q: “Composite shift?” — problem type, source and labeling all change MATH→GSM8K; reported as compound, single-factor decomposition planned.
- Sign convention: Δ = AUROC(embedding) – AUROC(TF-IDF), i.e. L = 1 – AUROC; Δ > 0 favors pretrained features.
- Split audit: 911,867 = 632,422 train + 88,872 val (reserved, unused) + 190,573 test — grouped by problem.
- AI use (§4.8): implementation code written by AI assistants under our direction; experimental design, data checking and ablation design by the group — we have read all the code and can explain every part.